Project 1: Co-creativity as Curation

MovieLens 100K Ranking & Personalization Pipeline

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Abstract

This report presents a co-creative curation pipeline built on the MovieLens 100K dataset. We implement and compare four ranking methods (a Popularity baseline, Matrix Factorization with Stochastic Gradient Descent (MF-SGD), Matrix Factorization with Alternating Least Squares (MF-ALS), and a Pairwise Learning-to-Rank (LTR) model) evaluated through Recall@10 and NDCG@10. Hyperparameters for the MF models were tuned using the Optuna framework. We also apply Maximal Marginal Relevance (MMR) reranking with cosine similarity based on genre to control list-level diversity, demonstrating a measurable relevance/diversity tradeoff across $\alpha \in \{0.1, 0.4, 0.7\}$. An ablation study isolates the contributions of MMR reranking and the popularity feature in LTR. Finally, we simulate three interactive personalization sessions using Exponential Moving Average (EMA) updates to observe how recommended lists evolve over consecutive feedback rounds.

1 Introduction

In modern entertainment platforms, the dominant creative act is often not the generation of a single artifact, but rather the curation, sequencing, and contextualization of what users experience next. This project explores this paradigm by framing curation as a form of co-creation. When a system recommends content, it applies selection pressure that ultimately shapes user tastes, creator incentives, and the visibility of different cultural niches.

Historically, recommender systems focused heavily on optimizing for prediction accuracy, such as minimizing the Root Mean Square Error (RMSE) of user ratings. However, optimizing solely for accuracy often leads to degenerate system behaviors, such as filter bubbles, popularity bias, and a lack of discovery. A truly co-creative pipeline must balance immediate relevance with list-level properties like diversity and pacing, while continuously adapting to user feedback.

To demonstrate this, we implemented an end-to-end interactive curation pipeline using the MovieLens 100K dataset. Our system blueprint is divided into four distinct stages:

1. **Candidate Generation:** Filtering the dataset to produce a viable pool of unseen movies for a given user.
2. **Ranking:** Scoring candidates to encode personal value and relevance. We compare four rankers: a Popularity baseline, Matrix Factorization trained via Stochastic Gradient Descent (MF-SGD), Matrix Factorization trained via Alternating Least Squares (MF-ALS), and a Pairwise Learning-to-Rank (LTR) model.

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3. **Reranking:** Applying Maximal Marginal Relevance (MMR) to the top-scored items to enforce diversity constraints, which makes the final recommended list explore different genres rather than collapsing into a narrow niche.

4. **Personalization Update:** Simulating interactive user sessions where the user's latent taste profile is dynamically updated after each choice using an Exponential Moving Average (EMA).

Through this pipeline, this report investigates the comparative offline performance of these rankers (Recall@10 and NDCG@10), analyzes the critical tradeoff between relevance and diversity, and observes how a user's state evolves during simulated co-creative sessions.

2 Dataset and Preprocessing

For this project, we utilized the **MovieLens 100K** dataset, a standard benchmark in recommender systems research that contains 100,000 ratings (ranging from 1 to 5) applied to 1,682 movies by 943 users.
To prepare the data for our models, we performed the following preprocessing steps:

1. **ID Normalization:** The original `user_id` and `item_id` values were mapped to contiguous integer indices starting from 0 (e.g., $u \in \{0, \ldots, 942\}$ and $i \in \{0, \ldots, 1681\}$). This allows us to use the indices directly for fast array lookups and matrix operations during training.

2. **Interaction Histories:** We built per-user and per-item interaction dictionaries. We also calculated the global mean rating (μ) across the training set, as well as the absolute popularity of each item (interaction count), which serves as a baseline feature for our rankers.

3. **Data Splitting:** We applied a randomized **80/20 train/test split** over the observed ratings. Additionally, to ease hyperparameter tuning without leaking test data, we applied a secondary 80/20 split on the training data to generate a dedicated **validation set**.

4. **Relevance Definition:** For the purposes of evaluating our Top-K recommendation lists, we defined the set of "relevant" test items for a user u as:

$$G_u = \{i : (u, i) \in \Omega_{\text{test}} \text{ and } r_{ui} \geq 4\}$$

where Ω_{test} represents the test set and r_{ui} is the true rating given by the user.

3 Implementation Details

The co-creative system is structured into four distinct stages: candidate generation, ranking, applying list constraints (reranking), and a personalization update via user feedback.

3.1 Candidate Generation

For any given user u, the candidate set $C(u)$ consists of all items that the user has not interacted with in the training split, to make sure our evaluation reflects the system's ability to recommend *unseen* content.

3.2 Rankers

We implemented four distinct ranking methods to encode personal relevance:

- **Popularity Baseline:** A non-personalized ranker that orders candidate movies based on their global interaction count in the training dataset.
- **MF-SGD:** A Matrix Factorization model with biases predicting ratings as $\hat{r}_{ui} = \mu + b_u + b_i + p_u^\top q_i$, trained using Stochastic Gradient Descent with separate regularization coefficients for latent factors and biases (see Section 3.2.1).
- **MF-ALS:** A Matrix Factorization model with biases trained using Alternating Least Squares, which relies on closed-form ridge regression updates derived in Appendix A.
- **Pairwise LTR:** A logistic pairwise preference model trained via gradient descent. The ranking score is $s_\phi(u,i) = \phi^\top f(u,i)$, where the feature vector is $f(u,i) = [s_{\mathrm{MF}}(u,i),\ b_u,\ b_i,\ \mathrm{pop}(i),\ 1]$, combining the MF score, user and item biases, log-normalized popularity, and a constant intercept term. Preference pairs (u, i^+, i^-) are constructed from training ratings with $r_{ui^+} \geq 4$ and $r_{ui^-} \leq 2$, capped at 50 pairs per user.

3.2.1 Hyperparameter Tuning

Rather than relying on manually chosen baseline parameters, we employed the `Optuna` framework to automatically optimize both MF models against the dedicated validation set by minimizing RMSE. This produced the following tuned configurations:

- **MF-SGD:** latent dimension $d = 81$, factor regularization $\lambda_{\text{factors}} \approx 0.1138$, bias regularization $\lambda_{\text{biases}} \approx 0.0005$, learning rate $\eta \approx 0.0116$, trained for 75 epochs (best checkpoint at epoch 56).
- **MF-ALS:** latent dimension $d = 131$, regularization $\lambda \approx 9.716$, 15 iterations (best checkpoint at iteration 5).

A key deviation from the standard practice lab implementation is that our SGD training uses **two separate regularization coefficients** instead of one. The standard formulation applies a single λ uniformly to all parameters:

$$l_{ui} = (r_{ui} - \hat{r}_{ui})^2 + \lambda\big(\|p_u\|^2 + \|q_i\|^2 + b_u^2 + b_i^2\big)$$

Our implementation decouples this into λ_{factors} for the latent vectors and λ_{biases} for the scalar bias terms:

$$l_{ui} = (r_{ui} - \hat{r}_{ui})^2 + \lambda_{\text{factors}}\big(\|p_u\|^2 + \|q_i\|^2\big) + \lambda_{\text{biases}}\big(b_u^2 + b_i^2\big)$$

This is motivated by the observation that bias terms are low-complexity scalar parameters that require much lighter regularization than the high-dimensional factor vectors. The Optuna study confirmed this: the optimal $\lambda_{\text{biases}} \approx 0.0005$ was more than two orders of magnitude smaller than $\lambda_{\text{factors}} \approx 0.1138$, validating the two-parameter design.

3.3 Reranking (Diversity Constraints)

To control list-level diversity, we applied Maximal Marginal Relevance (MMR) over the top $M = 80$ candidates produced by the base ranker. The reranker dynamically builds the slate step-by-step by selecting the item c^* that maximizes:

$$c^* = \arg\max_{c \in C \setminus S} \left((1-\alpha)\,\mathrm{rel}(c) - \alpha \max_{s \in S} \mathrm{sim}(c,s) \right)$$

where $\mathrm{rel}(c)$ is the normalized relevance score of candidate c and $\mathrm{sim}(c, s)$ is the cosine similarity between the L_2-normalized genre vectors of items c and s, extracted from the genre one-hot encoding (19 dimensions) in the dataset's item metadata. We explored the trade-off parameter $\alpha \in \{0.1, 0.4, 0.7\}$ to observe the impact on relevance and diversity metrics.

3.4 Personalization Update (Session)

To simulate a co-creative loop, the system executes multi-turn sessions where a user is presented with a ranked slate, makes a choice, and has their latent state updated. The simulated feedback heuristic selects whichever item in the top-10 has the highest known test rating, falling back to a random pick if no test ratings are present. The personalization update applies an Exponential Moving Average (EMA) to the user's latent representation. If user u selects item i^*, their factor vector p_u is updated as:

$$p_u \leftarrow (1 - \beta)\, p_u + \beta\, q_{i^*}$$

with $\beta = 0.1$. We simulated 5 consecutive feedback rounds for 3 distinct users to observe how the recommended slate evolves and adapts to recent choices.
The three users were selected by a principled, reproducible criterion: we ranked all eligible users (those with at least 2 relevant test items) by their number of relevant test items and selected one *active* user (most relevant items), one *median* user (median relevant items), and one *marginal* user (fewest relevant items while meeting the minimum threshold). This spread allows us to observe session dynamics across different levels of test-set signal.

4 Experiments and Results

4.1 Baseline Comparison

Table 1 reports Recall@10 and NDCG@10 averaged over 300 eligible users (those with $|G_u| > 0$), evaluated across all four rankers. Figure 1 visualises the same results as a grouped bar chart.

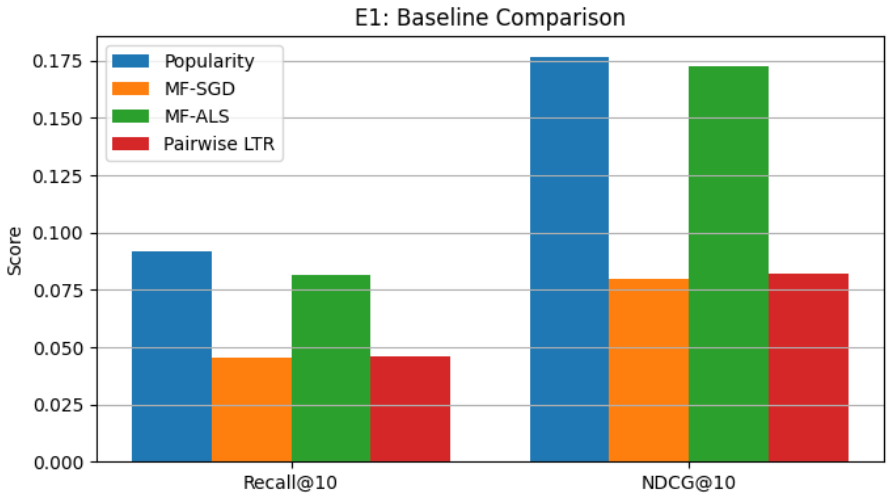

Figure 1: Grouped bar chart of Recall@10 and NDCG@10 for the four rankers.

Table 1: Offline ranking metrics for the four rankers ($K = 10$, 300 users).

Ranker	**Recall@10**	**NDCG@10**
Popularity	0.0917	0.1768
MF-SGD	0.0453	0.0799
MF-ALS	0.0816	0.1725
Pairwise LTR	0.0457	0.0823

The most striking result is that the non-personalized Popularity ranker achieves the highest scores on both metrics (Recall@10 = 0.0917, NDCG@10 = 0.1768) with MF-ALS close behind (0.0816, 0.1725), while MF-SGD and Pairwise LTR score substantially lower. This apparent paradox is a concrete manifestation of the offline evaluation caveat discussed in Appendix A (T5): the 80/20 random split produces a test set heavily skewed towards popular films, since widely-rated movies appear in many users' held-out sets. A ranker that systematically recommends the most-interacted items therefore achieves disproportionately high offline coverage.

A second notable finding is the inversion between RMSE and ranking quality for the two MF variants. MF-SGD achieves a lower validation RMSE (0.9246 vs. 0.9296 for MF-ALS), yet it ranks a full 0.0926 NDCG points below MF-ALS. This is a direct empirical demonstration of T4: a pointwise rating-prediction objective does not optimise item ordering. The SGD model's aggressive factor regularisation ($\lambda_{\text{factors}} \approx 0.114$) suppresses the magnitude of latent factor vectors and, consequently, the discriminative power of the inner-product score, whereas ALS, with its closed-form updates and strong ridge penalty ($\lambda = 9.716$), produces factor vectors that better preserve the global ordering of item appeal.

The Pairwise LTR model improves marginally over MF-SGD (NDCG: 0.0823 vs. 0.0799) despite using MF-SGD scores as its primary feature. Inspecting the learned weight vector reveals the mechanism: $\phi = [1.333,\ 0.000,\ 0.401,\ 0.046,\ 0.000]$. The user bias b_u and the intercept are driven to zero by the regulariser, which is mathematically expected: for a fixed user, b_u shifts all item scores by the same constant and thus has no effect on pairwise ordering. The item bias b_i ($\phi_2 = 0.401$) captures average item quality, and the log-popularity term ($\phi_3 = 0.046$) provides a small additional boost, together producing the marginal gain over the raw MF score.

4.2 MMR Relevance/Diversity Tradeoff

Table 2 and Figure 2 show the effect of the MMR trade-off parameter α on NDCG@10 and Diversity@10 when applied over the top-$M = 80$ candidates of the MF-SGD ranker. Diversity@10 is the average pairwise cosine distance between the L_2-normalised genre vectors of the 10 recommended items, evaluated over 200 users.

Table 2: MMR Relevance/Diversity Tradeoff: NDCG@10 and Diversity@10 across the full α range (MF-SGD base ranker, $K = 10$, 200 users).

α	**NDCG@10**	**Diversity@10**
0.0 (no MMR)	0.0817	0.7370
0.1	0.0804	0.7964
0.4	0.0688	0.9312
0.7	0.0667	0.9615
1.0 (pure diversity)	0.0667	0.9695

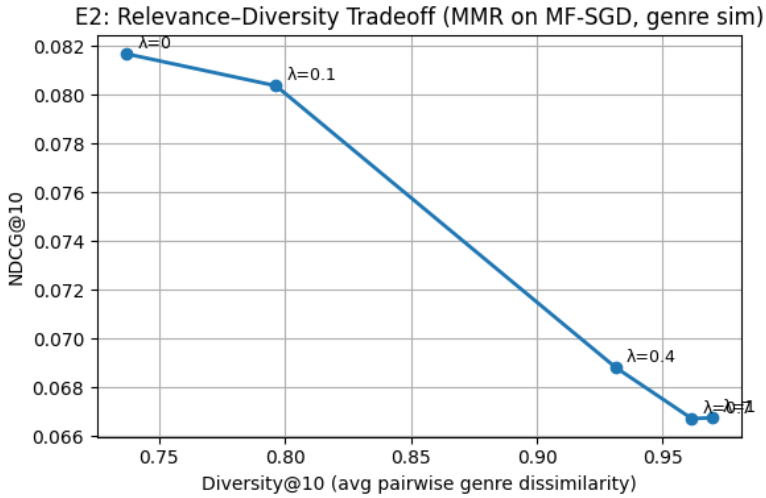

Figure 2: MMR Relevance/Diversity Tradeoff: NDCG@10 vs. Diversity@10 tradeoff curve across $\alpha \in \{0, 0.1, 0.4, 0.7, 1\}$.

The full tradeoff curve in spans from pure relevance to pure diversity, and is monotone throughout: every increase in α raises genre diversity while reducing or holding NDCG@10.
At the two extremes, $\alpha = 0$ corresponds to taking the top-K items directly from the MF-SGD ranking with no reranking (NDCG = 0.0817, Diversity = 0.7370), while $\alpha = 1$ ignores relevance entirely and selects greedily for genre dissimilarity (NDCG = 0.0667, Diversity = 0.9695). Notably, NDCG at $\alpha = 1$ is identical to that at $\alpha = 0.7$ to four decimal places, indicating that beyond $\alpha = 0.7$ the candidate pool has been sufficiently diversified and any further increase in the diversity weight displaces equally relevant items without gaining meaningful additional genre coverage.
The gains along the curve are strongly non-uniform. The step from $\alpha = 0$ to $\alpha = 0.1$ delivers a diversity gain of +0.059 at a negligible NDCG cost of only −0.0013 — a near-free improvement, as the first diversifying swaps replace genre near-duplicates sitting just below the top-ranked items. The largest single diversity jump occurs between $\alpha = 0.1$ and $\alpha = 0.4$ (+0.135 diversity, −0.0116 NDCG), where the reranker begins displacing genuinely relevant but genre-redundant items. The transition from $\alpha = 0.4$ to $\alpha = 0.7$ yields sharply diminishing returns (+0.030 diversity, −0.0021 NDCG), and the final step to $\alpha = 1$ adds only +0.008 diversity with no additional NDCG cost. This concave shape reflects the structure of the candidate pool: genre diversity is cheapest to gain early, when near-duplicates are available, and increasingly expensive as the remaining candidates become more heterogeneous. For the interactive sessions, $\alpha = 0.4$ was chosen as the default, capturing most of the available diversity gain (+0.194 over $\alpha = 0$) at a moderate and well-understood relevance cost.

4.3 Ablation Study

Two targeted ablations isolate the contribution of MMR reranking and the popularity feature in LTR.

Ablation A: Effect of MMR reranking. Table 3 compares the Pairwise LTR ranker with and without the MMR step ($\alpha = 0.4$, $K = 10$).

Table 3: Ablation A: Effect of MMR reranking on Pairwise LTR ($\alpha = 0.4$, $K = 10$).

Configuration	**NDCG@10**	**Recall@10**	**Diversity@10**
LTR (no MMR)	0.0823	0.0457	0.7539
LTR + MMR ($\alpha = 0.4$)	0.0682	0.0375	0.9359

Disabling MMR recovers 0.0141 NDCG and 0.0082 Recall but drops Diversity from 0.9359 to 0.7539, a loss of 0.1820 points. Without MMR, the LTR scorer tends to surface items from the same high-scoring genres (primarily Drama and Thriller), producing a slate that is internally redundant. MMR resolves this by explicitly penalising genre similarity to already-selected items, forcing the reranker to traverse a wider portion of the genre space at each step. The cost in ranking quality is real but bounded: NDCG falls by 17% while diversity improves by 24%, a tradeoff that is acceptable whenever list-level diversity is a system objective.

Ablation B: Effect of the popularity feature. Table 4 compares two LTR variants: the full 5-feature model $f = [s_{\mathrm{MF}}, b_u, b_i, \mathrm{pop}, 1]$ and a retrained 4-feature model with popularity removed, $f = [s_{\mathrm{MF}}, b_u, b_i, 1]$.

Table 4: Ablation B: Effect of removing the popularity feature from Pairwise LTR ($K = 10$).

Configuration	**NDCG@10**	**Recall@10**
LTR (full, 5 features)	0.0823	0.0457
LTR (no pop, 4 features)	0.0785	0.0441

Removing popularity reduces NDCG by 0.0038 (4.6%) and Recall by 0.0016 (3.5%). The no-pop weight vector $\phi_{\text{no-pop}} = [1.334,\ 0.000,\ 0.403,\ 0.000]$ is nearly identical to the full model on the shared features, confirming that the remaining features partially compensate but cannot fully substitute for the popularity signal. The drop, while modest, is consistent: popularity acts as a soft correction for exposure bias in the training set. Items that have accumulated many ratings are inherently more likely to appear in test sets drawn from the same distribution, so the ranker that acknowledges this signal recovers a small but reliable offline improvement.

4.4 Interactive Sessions

Table 5 reports NDCG@10 per turn for the three sessions, each running 5 rounds with LTR + MMR ($\alpha = 0.4$) and EMA updates ($\beta = 0.1$). Figure 3 plots the same data as a line chart. The three users were selected by a principled criterion: one active user (u=449, 95 relevant test items), one median user (u=387, 8 relevant test items), and one marginal user (u=432, 2 relevant test items), spanning the full range of test-set density.

Table 5: NDCG@10 per turn across the three user sessions (LTR + MMR, $\alpha = 0.4$, $\beta = 0.1$, $K = 10$).

Session (user id)	**Turn 1**	**Turn 2**	**Turn 3**	**Turn 4**	**Turn 5**
Active (u=449, 95 items)	0.2588	0.2985	0.1428	0.1396	0.2934
Median (u=387, 8 items)	0.0000	0.0761	0.0000	0.0000	0.0000
Marginal (u=432, 2 items)	0.1934	0.0000	0.0000	0.0000	0.0000

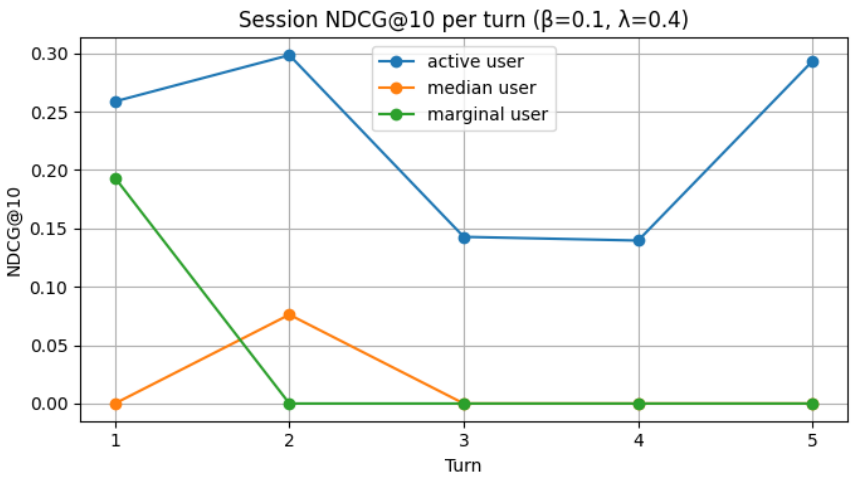

Figure 3: NDCG@10 per turn for the three user sessions ($\alpha = 0.4$, $\beta = 0.1$).

The three sessions exhibit qualitatively different dynamics, each informative in its own way (its logs are present in the delivery).

The **active user** achieves consistently high but non-linear NDCG across turns (0.259 → 0.299 → 0.143 → 0.140 → 0.293), which reflects two competing forces. On one hand, the EMA update ($\beta = 0.1$) progressively shifts the user embedding toward the latent space of chosen films; on the other, items shown in earlier turns are excluded from later candidate pools, narrowing the field. After three consecutive picks of critically acclaimed dramas and thrillers (Rear Window, Schindler's List, Psycho), the user embedding drifts towards that genre cluster and the candidate diversity drops (Diversity@10 falls to 0.709 in turn 4). Turn 5 recovers strongly (NDCG = 0.293) as the model rediscovers relevant items in adjacent parts of the embedding space. All five picks received a 5/5 test rating, confirming that the feedback heuristic surfaces genuinely preferred content.

The **median user** registers NDCG = 0 in four of five turns because the heuristic picks items not present in their test set (Casablanca, Wings of Desire, Henry V, As Good As It Gets). Only turn 2 yields a positive signal, when Titanic is recommended and rated 5/5 (NDCG = 0.0761, Recall = 0.1250, one of eight relevant items). This pattern illustrates a structural limitation of offline evaluation: the feedback heuristic can only confirm quality when the chosen item happens to appear in the held-out test ratings. Picks that are genuinely aligned with the user's taste but were not rated in the observed data register as failures.

The **marginal user** immediately surfaces *The Usual Suspects*, which the user rated 5/5 (NDCG = 0.1934, Recall = 0.5000, capturing one of only two relevant test items). Subsequent turns pick a stylistically coherent sequence of pre-1960 classics (The Third Man, The African Queen, My Man Godfrey, Arsenic and Old Lace), tracing a trajectory consistent with the noir/classic genre signal embedded in the first pick via the EMA update. However, with both relevant test items exhausted, offline NDCG cannot confirm the quality of these later recommendations, even though they reflect meaningful personalization.

5 Conclusion

The results of this project yield a few takeaways. First, regarding ranker quality, Popularity and MF-ALS dominate offline metrics (NDCG@10 of 0.1768 and 0.1725 respectively), while MF-SGD and Pairwise LTR trail significantly (0.0799 and 0.0823). This is primarily an artefact of offline evaluation bias rather than a failure of the personalized models: popular items are over-represented in held-out test sets drawn from the same distribution as training data, systematically rewarding rankers that promote them. The inversion of MF-SGD and MF-ALS relative

to their RMSE ordering (0.9246 vs. 0.9296) is a concrete illustration of why rating-prediction accuracy is a poor predictor of ranking quality. The Pairwise LTR model provides a marginal but consistent gain over MF-SGD by incorporating item bias and popularity as complementary features, and its learned weight vector reveals that user bias is correctly zeroed out for pairwise ranking.

Second, regarding the relevance–diversity tradeoff, MMR reranking with genre-vector similarity produces a clear, controllable tradeoff. The diversity gain at $\alpha = 0.1$ is nearly cost-free (-0.0013 NDCG, $+0.059$ Diversity), while $\alpha = 0.4$ offers a substantial diversity improvement ($+0.194$) at a 17% NDCG cost. For a co-creative system, concentrated slates accelerate filter-bubble formation, whereas diverse slates expose users to content they would not have self-selected, supporting discovery.

Third, regarding personalization dynamics, the EMA update shows that even a lightweight state-update rule produces qualitatively meaningful session trajectories. The active user's embedding drifts coherently toward their chosen genre cluster, and the recovery in turn 5 shows the model can escape temporary local concentration. The marginal user session further illustrates that meaningful personalization can occur even when offline metrics are silent, as the subsequent picks form a stylistically consistent sequence that reflects the embedding shift induced by the first choice.

A persistent limitation across all experiments is the offline evaluation framework itself: test ratings capture only items that users were historically exposed to, conflating exposure with preference. Future work could include debiasing via inverse propensity scoring, and replacing simulated feedback with real user interaction data to validate the session dynamics observed here under controlled conditions.

A Theory Questions

1. Stochastic Gradient Descent (SGD)

We consider the biased matrix factorization model

$$
\begin{aligned}
\mathcal{E}_{ui} &= R_{ui} - \hat{R}_{ui} \quad \text{(prediction error)} \\
\hat{R}_{ui} &= \mu + b_u + b_i + p_u^T q_i \quad \text{(prediction model)} \\
l_{ui} &= (R_{ui} - \hat{R}_{ui})^2 + \lambda(||p_u||^2 + ||q_i||^2 + b_u^2 + b_i^2) \quad \text{(regularized squared loss)}
\end{aligned}
$$

The SGD update rule for each parameter θ is

$$\theta \leftarrow \theta - \eta \nabla_\theta l_{ui},$$

where η denotes the learning rate.

Using the chain rule,

$$
\begin{aligned}
\frac{\partial}{\partial \theta}(R_{ui} - \hat{R}_{ui})^2 &= 2(R_{ui} - \hat{R}_{ui}) \cdot \frac{\partial}{\partial \theta}(R_{ui} - \hat{R}_{ui}) \\
&= -2\mathcal{E}_{ui} \cdot \frac{\partial \hat{R}_{ui}}{\partial \theta}.
\end{aligned}
$$

Gradient with respect to p_u Only the inner-product term depends on p_u, therefore

$$\nabla_{p_u} l_{ui} = -2\mathcal{E}_{ui} q_i + 2\lambda p_u.$$

Thus the SGD update becomes

$$p_u \leftarrow p_u + \eta(\mathcal{E}_{ui} q_i - \lambda p_u).$$

Gradient with respect to q_i Similarly,

$$\nabla_{q_i} l_{ui} = -2\mathcal{E}_{ui} p_u + 2\lambda q_i,$$

leading to

$$q_i \leftarrow q_i + \eta(\mathcal{E}_{ui} p_u - \lambda q_i).$$

Gradient with respect to b_u Since the prediction depends linearly on b_u,

$$\nabla_{b_u} l_{ui} = -2\mathcal{E}_{ui} + 2\lambda b_u,$$

and therefore

$$b_u \leftarrow b_u + \eta(\mathcal{E}_{ui} - \lambda b_u).$$

Gradient with respect to b_i Analogously,

$$\nabla_{b_i} l_{ui} = -2\mathcal{E}_{ui} + 2\lambda b_i,$$

so that

$$b_i \leftarrow b_i + \eta(\mathcal{E}_{ui} - \lambda b_i).$$

These updates define the SGD optimization procedure applied independently to each observed rating pair (u, i).

2. Alternating Least Squares (ALS)

Assume Q, b_u, b_i, and μ are fixed. Define the bias-corrected residual

$$e_{ui} = R_{ui} - (\mu + b_u + b_i).$$

Then optimizing over p_u reduces to

$$L(p_u) = \sum_{i \in \Omega(u)} (e_{ui} - p_u^T q_i)^2 + \lambda ||p_u||^2,$$

which corresponds to a ridge regression problem with predictors q_i and targets e_{ui}.
Taking the gradient with respect to p_u,

$$\nabla_{p_u} L(p_u) = \sum_{i \in \Omega(u)} 2(e_{ui} - p_u^T q_i)(-q_i) + 2\lambda p_u.$$

Setting the gradient equal to zero yields the normal equations

$$\sum_{i \in \Omega(u)} (e_{ui} - p_u^T q_i)(-q_i) + \lambda p_u = 0.$$

Expanding terms,

$$\sum_{i \in \Omega(u)} e_{ui} q_i = \sum_{i \in \Omega(u)} (q_i q_i^T) p_u + \lambda p_u.$$

Factoring p_u,

$$\sum_{i \in \Omega(u)} e_{ui} q_i = \left(\sum_{i \in \Omega(u)} q_i q_i^T + \lambda I \right) p_u.$$

Multiplying both sides by the inverse matrix gives the closed-form solution

$$p_u = \left(\sum_{i \in \Omega(u)} q_i q_i^T + \lambda I \right)^{-1} \left(\sum_{i \in \Omega(u)} e_{ui} q_i \right).$$

This expression corresponds to the ALS update step for user latent factors under ℓ_2 regularization.

3. Pairwise gradient

Let

$$\begin{aligned} s_\phi(u, i) &= \phi^T f(u, i), \\ \Delta f &= f(u, i^+) - f(u, i^-). \end{aligned}$$

We want to compute

$$\nabla_\phi \left(-\log \sigma(\phi^T \Delta f) \right).$$

Define

$$z = \phi^T \Delta f.$$

Since

$$\sigma(z) = \frac{1}{1+e^{-z}},$$

its derivative is

$$\sigma'(z) = \sigma(z)(1-\sigma(z)).$$

Therefore,

$$\begin{aligned}\frac{d}{dz}\left(-\log\sigma(z)\right) &= -\frac{1}{\sigma(z)}\sigma'(z)\\ &= -(1-\sigma(z))\\ &= \sigma(z)-1.\end{aligned}$$

Using the chain rule,

$$\nabla_\phi z = \nabla_\phi(\phi^T\Delta f) = \Delta f.$$

Thus,

$$\nabla_\phi\left(-\log\sigma(\phi^T\Delta f)\right) = (\sigma(\phi^T\Delta f)-1)\Delta f.$$

4. Why RMSE $\neq$ ranking quality?

Optimizing RMSE in matrix factorization minimizes prediction error over rating values:

$$\text{RMSE} = \sqrt{\frac{1}{|\Omega|}\sum_{(u,i)\in\Omega}(R_{ui}-\hat{R}_{ui})^2}.$$

However, top-K recommendation quality depends primarily on the correctness of the ordering of predicted scores rather than their absolute values.
There are several reasons for this mismatch:

- RMSE penalizes magnitude errors instead of ranking errors. Small prediction changes can significantly affect ranking while barely changing RMSE.
- It treats all prediction errors equally, whereas recommendation quality depends mainly on correctly ranking the highest-scored items.
- The absence of a rating is typically related to its value. Users rate only items they are exposed to, introducing exposure bias into the dataset.
- It does not enforce pairwise ordering constraints such as $\hat{R}_{ui^+} > \hat{R}_{ui^-}$, which is essential for ranking-based recommendation tasks.

Thus, in general, low RMSE does not necessarily imply good top-K recommendation quality.

5. Offline evaluation caveat

Offline evaluation assumes that observed interactions reflect true user preferences. However, this assumption fails when exposure is not random.
We can define interaction as

$$\text{interaction} = \text{exposure} + \text{preference}.$$

Users can only interact with items they are exposed to. Therefore, interaction data reflects both user preferences and historical exposure policies.
As a consequence:

- Popular items are overrepresented in the dataset, leading to popularity bias.
- New or rarely exposed items appear artificially worse during evaluation.
- Offline evaluation lacks counterfactual information about how users would behave if exposed to different items.

Thus, offline evaluation can produce misleading estimates of recommendation quality unless exposure is randomized or corrected using debiasing techniques, such as inverse propensity scoring, in which data is reweighted to account for the probability of exposure.